Wine Quality Prediction Analysis

Combined Red and White Wine Regression Analysis

Your Name

2026-04-18

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1 Executive Summary

1.1 Overview

This report analyzes the **UCI Wine Quality** dataset to examine how physicochemical measurements relate to wine quality scores.1 Following the structure of the full analysis workflow, I merge the red and white wine files, add a binary wine-type indicator, explore the data, fit ordinary least squares regression models, evaluate multicollinearity and diagnostics, and then compare OLS with three shrinkage methods: Ridge, Lasso, and Elastic Net.

Because the UCI repository describes `quality` as a numeric score and explicitly notes that the dataset can be used for regression, a regression framework is an appropriate starting point for this project.1 The repository also includes the local file `data/raw/winequality.names`, which serves as the in-repo data dictionary alongside the raw CSV files.

1.2 Dataset

The analysis uses three raw files stored in the repository through relative paths:

File	Role in the project
`data/raw/winequality-red.csv`	Raw red wine observations
`data/raw/winequality-white.csv`	Raw white wine observations
`data/raw/winequality.names`	Local variable and dataset description

This structure keeps the report self-contained. The code reads the data directly from the repository, while the dataset citation points back to the public UCI source.1

2 Setup and Data Loading

2.1 Reproducibility Setup

2.2 Load Packages

2.3 Load and Merge Data

Table 2: Dimensions of the red, white, and combined wine datasets

dataset	rows	columns
Red wine	1599	13
White wine	4898	13
Combined	6497	14

The merged dataset allows me to estimate a pooled model while still retaining wine type as an explicit predictor. That makes it possible to measure overall effects and also check whether the white-versus-red distinction matters after controlling for the chemical variables.

2.4 Data Structure

```
'data.frame':   6497 obs. of  14 variables:
 $ fixed.acidity       : num  7.4 7.8 7.8 11.2 7.4 7.4 7.9 7.3 7.8 7.5 ...
 $ volatile.acidity    : num  0.7 0.88 0.76 0.28 0.7 0.66 0.6 0.65 0.58 0.5 ...
 $ citric.acid         : num  0 0 0.04 0.56 0 0 0.06 0 0.02 0.36 ...
 $ residual.sugar      : num  1.9 2.6 2.3 1.9 1.9 1.8 1.6 1.2 2 6.1 ...
 $ chlorides           : num  0.076 0.098 0.092 0.075 0.076 0.075 0.069 0.065 0.073 0.071 ..
 $ free.sulfur.dioxide : num  11 25 15 17 11 13 15 15 9 17 ...
 $ total.sulfur.dioxide: num  34 67 54 60 34 40 59 21 18 102 ...
 $ density             : num  0.998 0.997 0.997 0.998 0.998 ...
 $ pH                  : num  3.51 3.2 3.26 3.16 3.51 3.51 3.3 3.39 3.36 3.35 ...
 $ sulphates           : num  0.56 0.68 0.65 0.58 0.56 0.56 0.46 0.47 0.57 0.8 ...
 $ alcohol             : num  9.4 9.8 9.8 9.8 9.4 9.4 9.4 10 9.5 10.5 ...
 $ quality             : int  5 5 5 6 5 5 5 7 7 5 ...
 $ white               : num  0 0 0 0 0 0 0 0 0 0 ...
 $ wine_type           : Factor w/ 2 levels "Red","White": 1 1 1 1 1 1 1 1 1 1 ...
```

Table 3: Missing values by variable in the combined dataset

variable	missing_values
fixed.acidity	0
volatile.acidity	0
citric.acid	0
residual.sugar	0
chlorides	0
free.sulfur.dioxide	0
total.sulfur.dioxide	0
density	0
pH	0
sulphates	0
alcohol	0
quality	0
white	0
wine_type	0

The data are already numeric and clean enough for immediate modeling. This is useful because it lets the report focus on analysis rather than extensive preprocessing.1

3 Exploratory Data Analysis

3.1 Quality Distribution

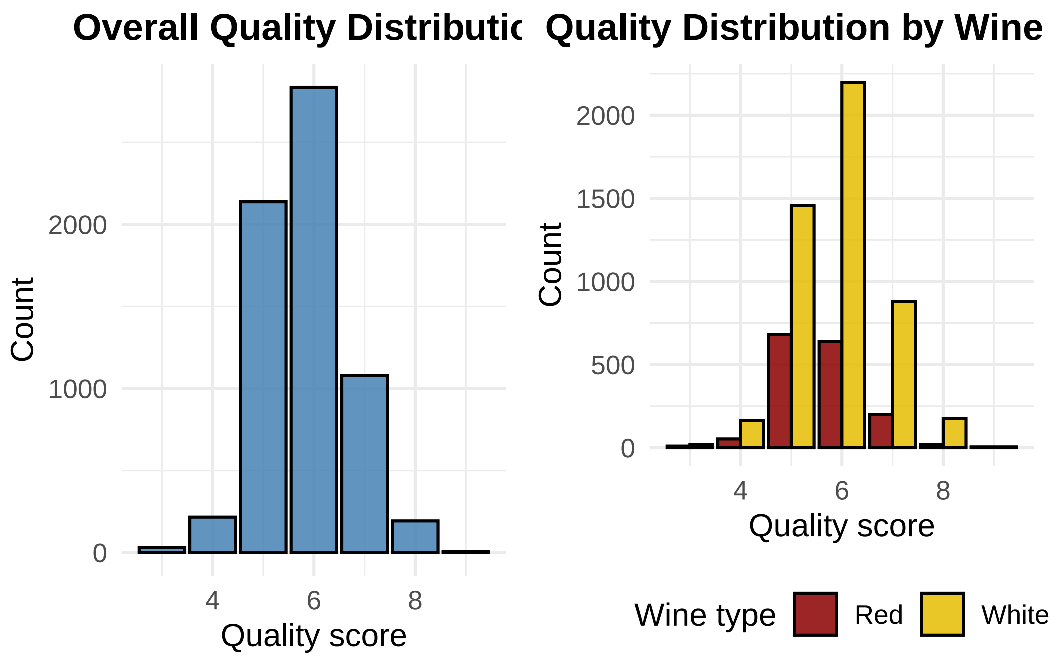

Figure 1: Overall and wine-type-specific distributions of wine quality

The quality distributions show that the response variable is concentrated in the middle score range rather than being evenly spread across all possible values. The grouped version also shows that red and white wines are not distributed identically, which reinforces the value of including a wine-type indicator in the pooled model.

3.2 Quality Statistics by Wine Type

Table 4: Descriptive statistics of quality by wine type

wine_type	n	mean_quality	sd_quality	median_quality	min_quality	max_quality
Red	1599	5.64	0.81	6	3	8
White	4898	5.88	0.89	6	3	9

This summary provides a direct comparison of the response variable across wine types. It helps establish whether mean quality and variability differ before moving into multivariable modeling.

3.3 Correlation Analysis

Table 5: Correlations between predictors and wine quality

predictor	correlation_with_quality
alcohol	0.444
density	-0.306
volatile.acidity	-0.266
chlorides	-0.201
white	0.119
citric.acid	0.086
fixed.acidity	-0.077
free.sulfur.dioxide	0.055
total.sulfur.dioxide	-0.041
sulphates	0.038
residual.sugar	-0.037
pH	0.020

The correlation table gives a first-pass view of the strongest bivariate relationships. It is not a substitute for regression, but it does help identify which variables are most likely to remain important in the multivariable models.

3.4 Highly Correlated Predictor Pairs

Table 6: Highly correlated predictor pairs in the combined dataset

var_1	var_2	correlation
total.sulfur.dioxide	free.sulfur.dioxide	0.721
white	total.sulfur.dioxide	0.700
alcohol	density	-0.687
white	volatile.acidity	-0.653

Strong correlations among predictors do not automatically invalidate OLS, but they do raise the possibility of multicollinearity. That is why the formal VIF check later in the report is an important complement to this exploratory step.

3.5 Alcohol and Quality

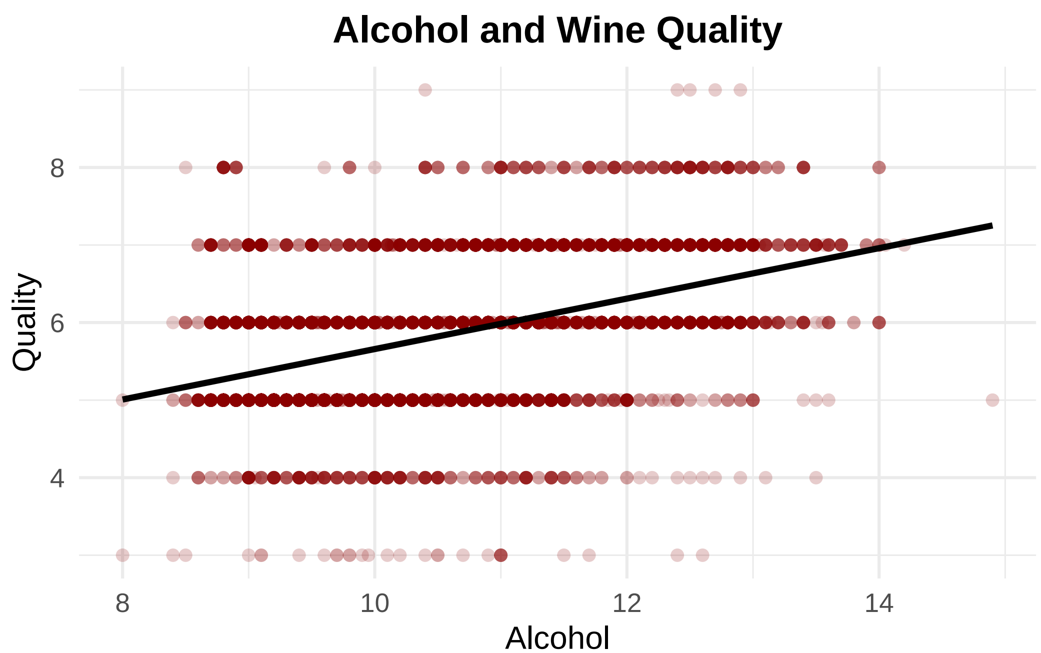

Figure 2: Relationship between alcohol content and wine quality

The alcohol plot provides a concrete example of a predictor-response relationship before the multivariable model adjusts for the rest of the chemical variables.

4 Traditional OLS Regression Pipeline

4.1 Full Model with All Predictors

```
Call:
lm(formula = quality ~ . - wine_type, data = wine_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.7796 -0.4671 -0.0444  0.4561  3.0211 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)           1.048e+02  1.414e+01   7.411 1.42e-13 ***
```

```
fixed.acidity         8.507e-02  1.576e-02   5.396 7.05e-08 ***
volatile.acidity     -1.492e+00  8.135e-02 -18.345  < 2e-16 ***
citric.acid          -6.262e-02  7.972e-02  -0.786   0.4322    
residual.sugar        6.244e-02  5.934e-03  10.522  < 2e-16 ***
chlorides            -7.573e-01  3.344e-01  -2.264   0.0236 *  
free.sulfur.dioxide   4.937e-03  7.662e-04   6.443 1.25e-10 ***
total.sulfur.dioxide -1.403e-03  3.237e-04  -4.333 1.49e-05 ***
density              -1.039e+02  1.434e+01  -7.248 4.71e-13 ***
pH                    4.988e-01  9.058e-02   5.506 3.81e-08 ***
sulphates             7.217e-01  7.624e-02   9.466  < 2e-16 ***
alcohol               2.227e-01  1.807e-02  12.320  < 2e-16 ***
white                -3.613e-01  5.675e-02  -6.367 2.06e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7331 on 6484 degrees of freedom
Multiple R-squared:  0.2965,    Adjusted R-squared:  0.2952 
F-statistic: 227.8 on 12 and 6484 DF,  p-value: < 2.2e-16
```

The full OLS model serves as the baseline specification. It estimates the partial association between each predictor and quality while holding the other variables constant.

Table 7: Statistically significant predictors in the full OLS model

predictor	estimate	std_error	t_value	p_value
fixed.acidity	0.0851	0.0158	5.3962	0.0000
volatile.acidity	-1.4924	0.0814	-18.3454	0.0000
residual.sugar	0.0624	0.0059	10.5218	0.0000
chlorides	-0.7573	0.3344	-2.2642	0.0236
free.sulfur.dioxide	0.0049	0.0008	6.4432	0.0000
total.sulfur.dioxide	-0.0014	0.0003	-4.3334	0.0000
density	-103.9096	14.3360	-7.2482	0.0000
pH	0.4988	0.0906	5.5063	0.0000
sulphates	0.7217	0.0762	9.4664	0.0000
alcohol	0.2227	0.0181	12.3199	0.0000
white	-0.3613	0.0568	-6.3667	0.0000

This table highlights the predictors that remain statistically associated with quality after adjustment for the rest of the model.

4.2 VIF Check for Multicollinearity

Table 8: Variance inflation factors for the full OLS model

predictor	vif
density	22.34
residual.sugar	9.63
white	7.22
alcohol	5.62
fixed.acidity	5.05
total.sulfur.dioxide	4.05
pH	2.56
free.sulfur.dioxide	2.24
volatile.acidity	2.17
chlorides	1.66
citric.acid	1.62
sulphates	1.56

Table 9: Predictors flagged by an informal VIF threshold greater than 5

predictor	vif
density	22.34
residual.sugar	9.63
white	7.22
alcohol	5.62
fixed.acidity	5.05

The VIF results help identify whether some coefficients are being inflated by predictor overlap. This is one reason the shrinkage methods later in the report are useful, since regularization can stabilize estimates when multicollinearity is present.

4.3 AIC Backward Selection

```
Start:  AIC=-4021.28
quality ~ (fixed.acidity + volatile.acidity + citric.acid + residual.sugar +
    chlorides + free.sulfur.dioxide + total.sulfur.dioxide +
    density + pH + sulphates + alcohol + white + wine_type) -
    wine_type
```

```
                       Df Sum of Sq    RSS     AIC
- citric.acid           1     0.332 3485.1 -4022.7
<none>                              3484.7 -4021.3
- chlorides             1     2.755 3487.5 -4018.1
- total.sulfur.dioxide  1    10.092 3494.8 -4004.5
- fixed.acidity         1    15.650 3500.4 -3994.2
- pH                    1    16.295 3501.0 -3993.0
- white                 1    21.785 3506.5 -3982.8
- free.sulfur.dioxide   1    22.312 3507.1 -3981.8
- density               1    28.235 3513.0 -3970.8
- sulphates             1    48.161 3532.9 -3934.1
- residual.sugar        1    59.499 3544.2 -3913.3
- alcohol               1    81.573 3566.3 -3872.9
- volatile.acidity      1   180.876 3665.6 -3694.5

Step:  AIC=-4022.66
quality ~ fixed.acidity + volatile.acidity + residual.sugar + 
    chlorides + free.sulfur.dioxide + total.sulfur.dioxide + 
    density + pH + sulphates + alcohol + white

                       Df Sum of Sq    RSS     AIC
<none>                              3485.1 -4022.7
- chlorides             1     3.168 3488.2 -4018.8
- total.sulfur.dioxide  1    10.537 3495.6 -4005.0
- fixed.acidity         1    15.390 3500.5 -3996.0
- pH                    1    16.775 3501.9 -3993.5
- free.sulfur.dioxide   1    22.351 3507.4 -3983.1
- white                 1    22.479 3507.6 -3982.9
- density               1    28.697 3513.8 -3971.4
- sulphates             1    47.876 3533.0 -3936.0
- residual.sugar        1    59.787 3544.9 -3914.1
- alcohol               1    81.509 3566.6 -3874.5
- volatile.acidity      1   197.707 3682.8 -3666.2


Call:
lm(formula = quality ~ fixed.acidity + volatile.acidity + residual.sugar + 
    chlorides + free.sulfur.dioxide + total.sulfur.dioxide + 
    density + pH + sulphates + alcohol + white, data = wine_data)

Residuals:
    Min      1Q  Median      3Q     Max 
```

```
-3.7708 -0.4696 -0.0418  0.4540  3.0202

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)           1.054e+02  1.411e+01   7.471 9.02e-14 ***
fixed.acidity         8.244e-02  1.541e-02   5.351 9.02e-08 ***
volatile.acidity     -1.471e+00  7.670e-02 -19.180  < 2e-16 ***
residual.sugar        6.256e-02  5.932e-03  10.548  < 2e-16 ***
chlorides            -8.007e-01  3.298e-01  -2.428   0.0152 *  
free.sulfur.dioxide   4.941e-03  7.662e-04   6.449 1.21e-10 ***
total.sulfur.dioxide -1.427e-03  3.222e-04  -4.428 9.66e-06 ***
density              -1.046e+02  1.431e+01  -7.307 3.05e-13 ***
pH                    5.044e-01  9.029e-02   5.587 2.40e-08 ***
sulphates             7.186e-01  7.614e-02   9.439  < 2e-16 ***
alcohol               2.210e-01  1.794e-02  12.315  < 2e-16 ***
white                -3.655e-01  5.651e-02  -6.468 1.07e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7331 on 6485 degrees of freedom
Multiple R-squared:  0.2965,    Adjusted R-squared:  0.2953 
F-statistic: 248.4 on 11 and 6485 DF,  p-value: < 2.2e-16
```

Table 10: Predictors retained and removed by backward AIC selection

status	predictor
Retained	fixed.acidity
Retained	volatile.acidity
Retained	residual.sugar
Retained	chlorides
Retained	free.sulfur.dioxide
Retained	total.sulfur.dioxide
Retained	density
Retained	pH
Retained	sulphates
Retained	alcohol
Retained	white
Removed	citric.acid

Backward AIC selection simplifies the baseline specification by dropping predictors that contribute less under the chosen information criterion. The selected OLS model becomes the main classical-regression benchmark for the rest of the report.

4.4 Reduced Model Coefficients

Table 11: Coefficient estimates for the AIC-selected OLS model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	105.4133	14.1100	7.4708	0.0000	77.7530	133.0736
fixed.acidity	0.0824	0.0154	5.3515	0.0000	0.0522	0.1126
volatile.acidity	-1.4711	0.0767	-19.1805	0.0000	-1.6215	-1.3208
residual.sugar	0.0626	0.0059	10.5475	0.0000	0.0509	0.0742
chlorides	-0.8007	0.3298	-2.4278	0.0152	-1.4473	-0.1542
free.sulfur.dioxide	0.0049	0.0008	6.4491	0.0000	0.0034	0.0064
total.sulfur.dioxide	-0.0014	0.0003	-4.4281	0.0000	-0.0021	-0.0008
density	-104.5741	14.3105	-7.3075	0.0000	-132.6275	-76.5207
pH	0.5044	0.0903	5.5871	0.0000	0.3274	0.6814
sulphates	0.7186	0.0761	9.4386	0.0000	0.5694	0.8679
alcohol	0.2210	0.0179	12.3155	0.0000	0.1858	0.2561
white	-0.3655	0.0565	-6.4675	0.0000	-0.4762	-0.2547

The coefficient table supports interpretation by showing both estimated effects and confidence intervals for the final OLS model.

4.5 OLS Model Diagnostics

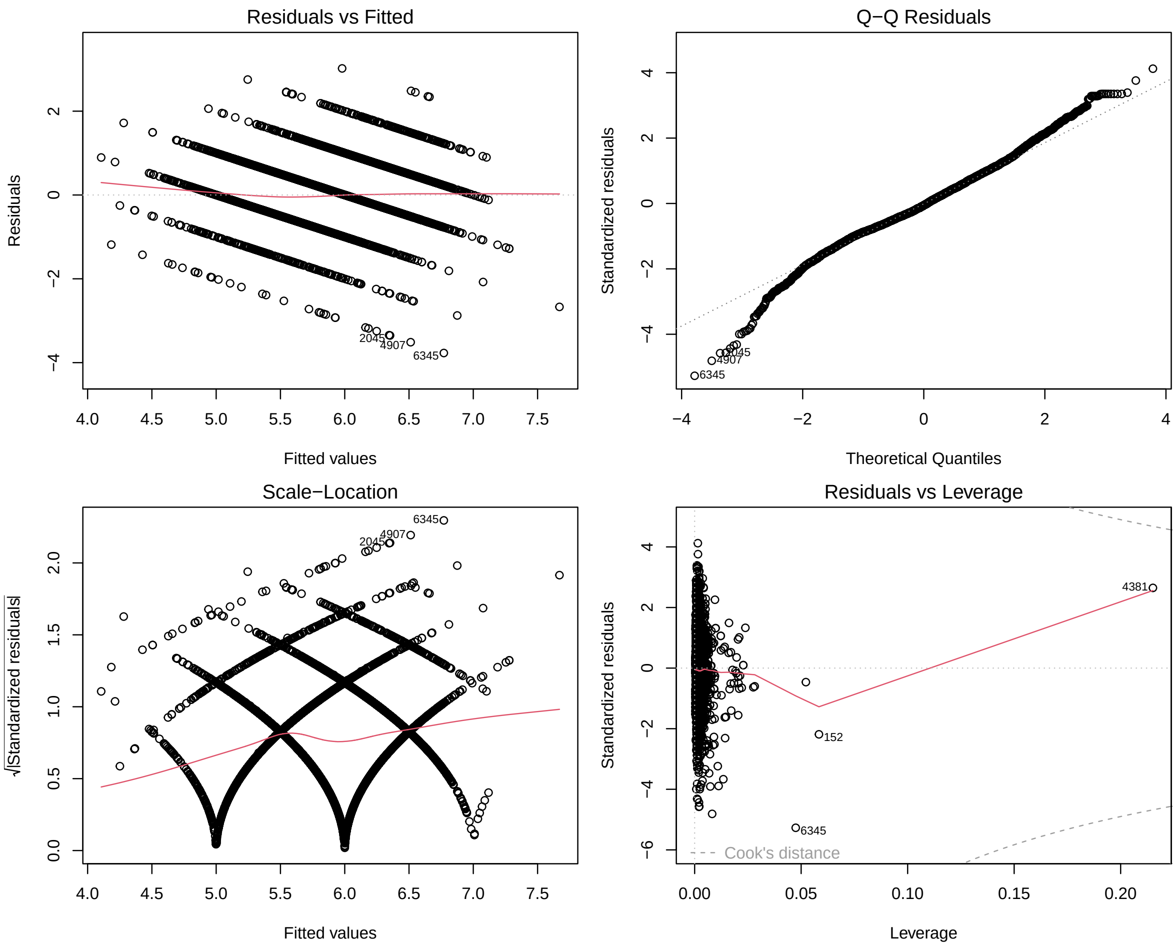

Figure 3: Diagnostic plots for the AIC-selected OLS model

Table 12: Formal diagnostic tests for the AIC-selected OLS model

test	statistic	p_value
Shapiro-Wilk normality	0.9882	0
Breusch-Pagan homoscedasticity	113.8934	0
Durbin-Watson independence	1.6472	0

The diagnostic section checks whether the selected OLS model shows obvious assumption problems. In a dataset of this size, small departures are possible, but the combined visual and formal checks still provide a useful benchmark before comparing OLS to shrinkage methods.

5 Shrinkage Methods

5.1 Data Preparation for `glmnet`

Table 13: Dimensions of the model matrix used for shrinkage methods

rows	columns	response_min	response_max
6497	12	3	9

The shrinkage models use the same merged dataset, but they require a numeric model matrix rather than a formula interface. This setup preserves comparability across OLS, Ridge, Lasso, and Elastic Net.

5.2 Ridge Regression

Ridge regression minimizes the penalized objective

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2,$$

which shrinks coefficients toward zero without setting them exactly equal to zero. This is especially useful when predictors are correlated.

Table 14: Ridge regression coefficients at the cross-validated optimal lambda

predictor	coefficient
density	-49.8105
volatile.acidity	-1.3725
chlorides	-1.0077
sulphates	0.6471
pH	0.2772
alcohol	0.2622
white	-0.1933
residual.sugar	0.0375
fixed.acidity	0.0367
citric.acid	-0.0144
free.sulfur.dioxide	0.0049
total.sulfur.dioxide	-0.0015

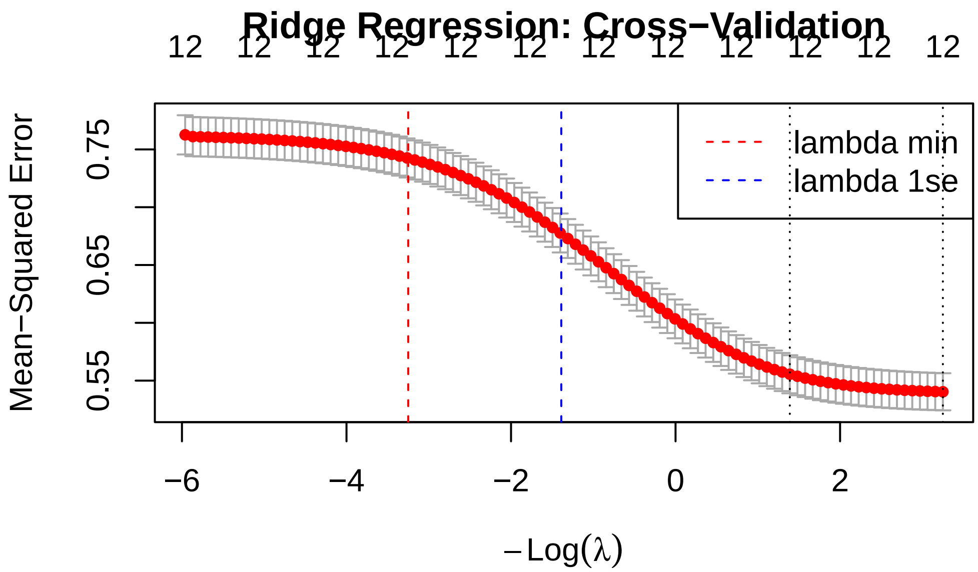

Figure 4: Cross-validation curve for Ridge regression

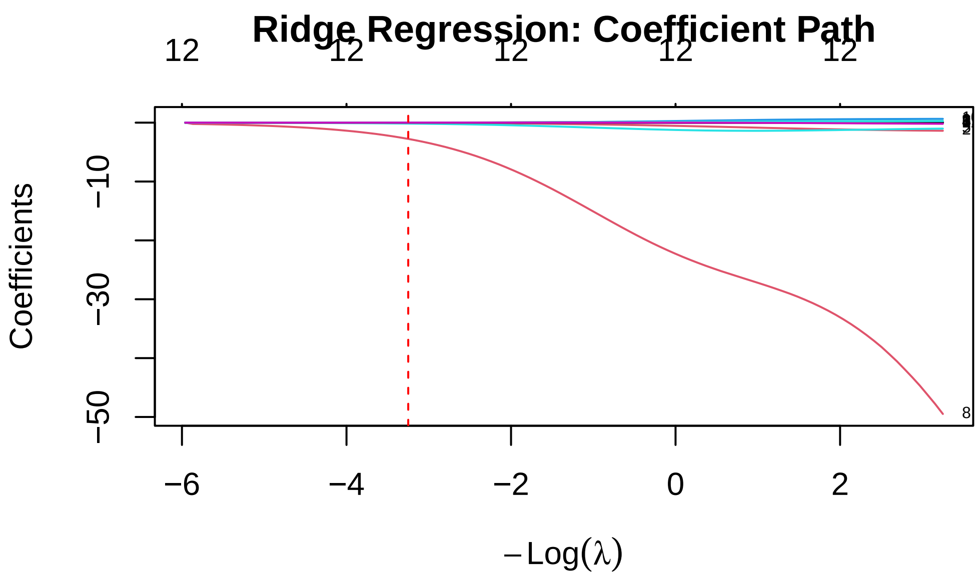

Figure 5: Coefficient path for Ridge regression

5.3 Lasso Regression

Lasso regression minimizes the penalized objective

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|,$$

which can shrink some coefficients exactly to zero. This makes Lasso useful when variable selection is part of the modeling goal.

Table 15: Lasso regression coefficients at the cross-validated optimal lambda

predictor	coefficient
density	-95.3933
volatile.acidity	-1.4890
chlorides	-0.7641
sulphates	0.7084
pH	0.4612
white	-0.3416
alcohol	0.2312
fixed.acidity	0.0769
residual.sugar	0.0589
citric.acid	-0.0569
free.sulfur.dioxide	0.0049
total.sulfur.dioxide	-0.0014

Table 16: Variables retained by Lasso

predictor	coefficient
density	-95.3933
volatile.acidity	-1.4890
chlorides	-0.7641
sulphates	0.7084
pH	0.4612
white	-0.3416
alcohol	0.2312
fixed.acidity	0.0769
residual.sugar	0.0589
citric.acid	-0.0569
free.sulfur.dioxide	0.0049
total.sulfur.dioxide	-0.0014

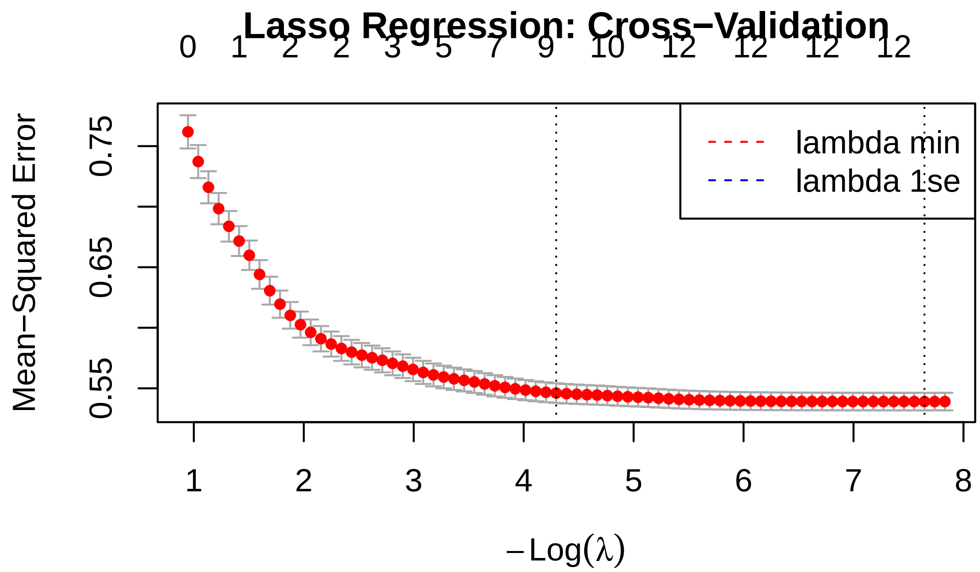

Figure 6: Cross-validation curve for Lasso regression

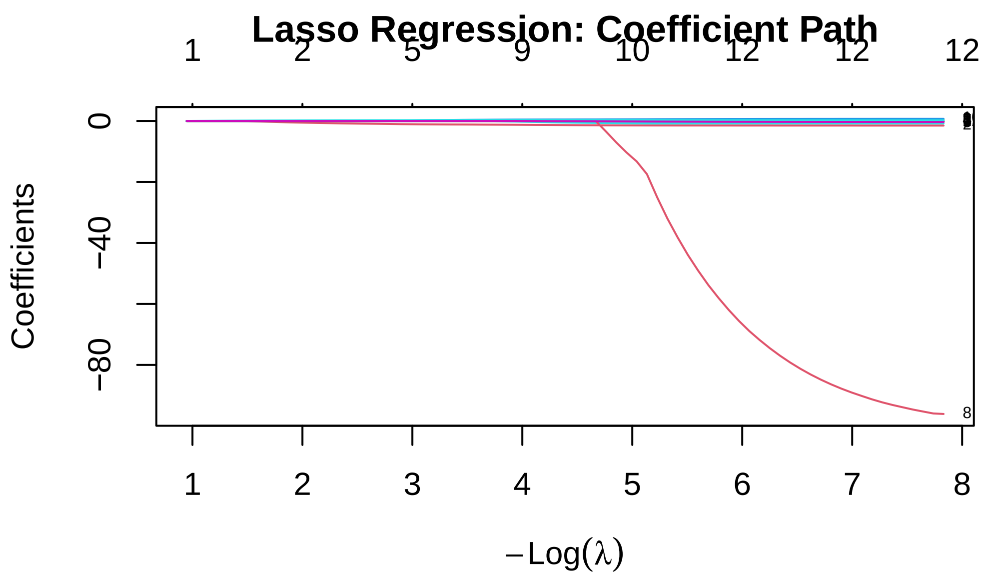

Figure 7: Coefficient path for Lasso regression

5.4 Elastic Net

Elastic Net combines the Ridge and Lasso penalties according to

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \sum_{j=1}^{p} \beta_j^2\right].$$

This method is useful when predictors are correlated but I still want some level of variable selection.

Table 17: Elastic Net coefficients at the selected alpha and lambda values

predictor	coefficient
density	-96.2779
volatile.acidity	-1.4896
chlorides	-0.7678
sulphates	0.7105
pH	0.4656
white	-0.3434
alcohol	0.2303
fixed.acidity	0.0779
residual.sugar	0.0593
citric.acid	-0.0585
free.sulfur.dioxide	0.0049
total.sulfur.dioxide	-0.0014

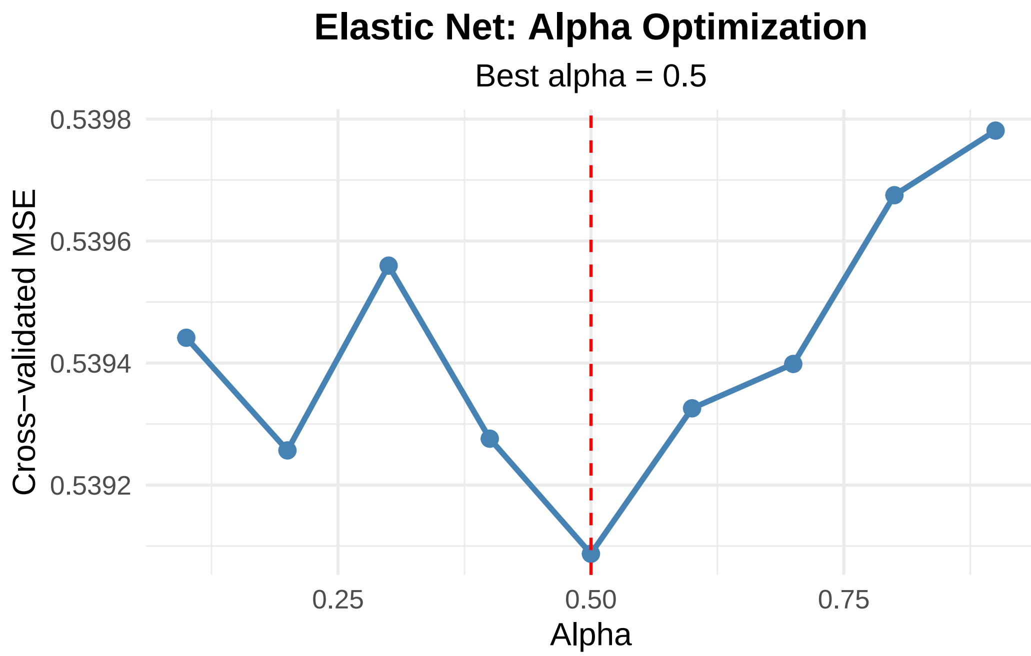

Figure 8: Cross-validated MSE across Elastic Net alpha values

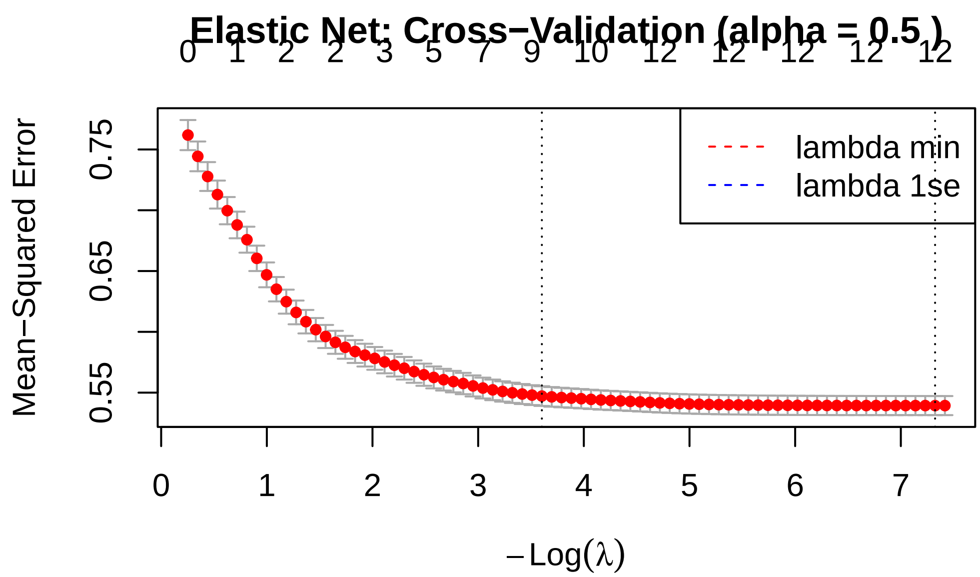

Figure 9: Cross-validation curve for Elastic Net

6 Model Comparison

6.1 Performance Metrics

Table 18: Performance metrics across the selected OLS, Ridge, Lasso, and Elastic Net models

model	r_squared	adjusted_r_squared	rmse	mae	predictors
OLS	0.2965	0.2953	0.7324	0.5686	11
Ridge	0.2938	0.2925	0.7338	0.5701	12
Lasso	0.2965	0.2952	0.7324	0.5687	12
Elastic Net	0.2965	0.2952	0.7324	0.5686	12

This table provides a direct side-by-side comparison of the competing modeling approaches. It shows whether shrinkage improves fit enough to justify the extra modeling complexity.

6.2 Visual Comparison of Model Performance

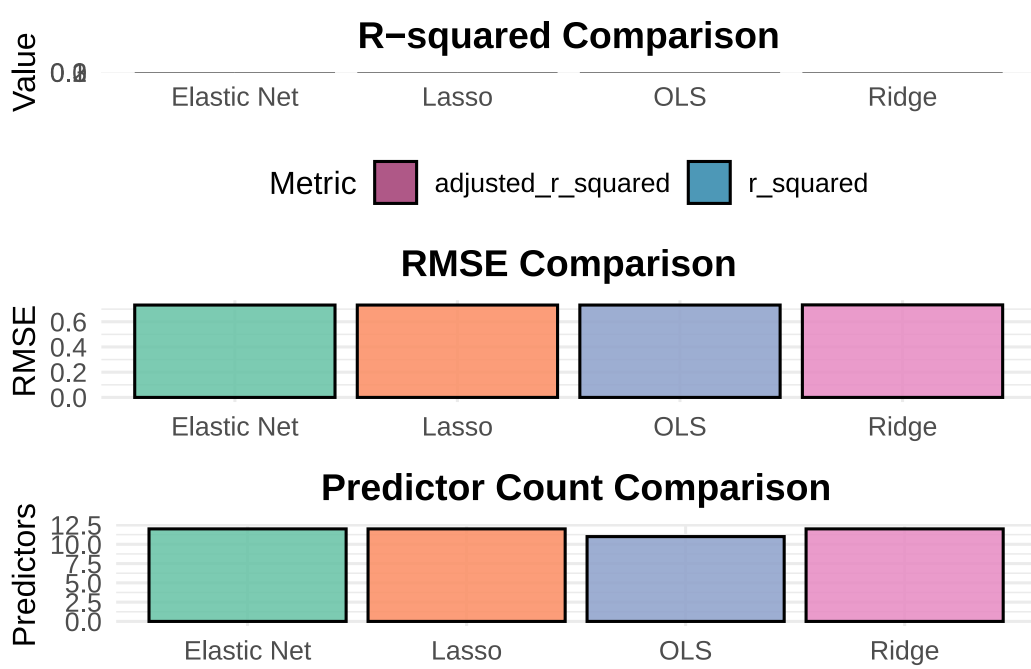

Figure 10: Visual comparison of model performance across OLS and shrinkage methods

The visual comparison makes it easier to see the trade-off between fit and complexity. In particular, it helps distinguish a model that predicts slightly better from a model that is simpler and easier to explain.

6.3 Coefficient Comparison

Table 19: Coefficient comparison across OLS, Ridge, Lasso, and Elastic Net

term	OLS	Ridge	Lasso	Elastic Net
fixed.acidity	0.0824	0.0367	0.0769	0.0779
volatile.acidity	-1.4711	-1.3725	-1.4890	-1.4896
residual.sugar	0.0626	0.0375	0.0589	0.0593
chlorides	-0.8007	-1.0077	-0.7641	-0.7678
free.sulfur.dioxide	0.0049	0.0049	0.0049	0.0049
total.sulfur.dioxide	-0.0014	-0.0015	-0.0014	-0.0014
density	-104.5741	-49.8105	-95.3933	-96.2779
pH	0.5044	0.2772	0.4612	0.4656
sulphates	0.7186	0.6471	0.7084	0.7105
alcohol	0.2210	0.2622	0.2312	0.2303
white	-0.3655	-0.1933	-0.3416	-0.3434
citric.acid	NA	-0.0144	-0.0569	-0.0585

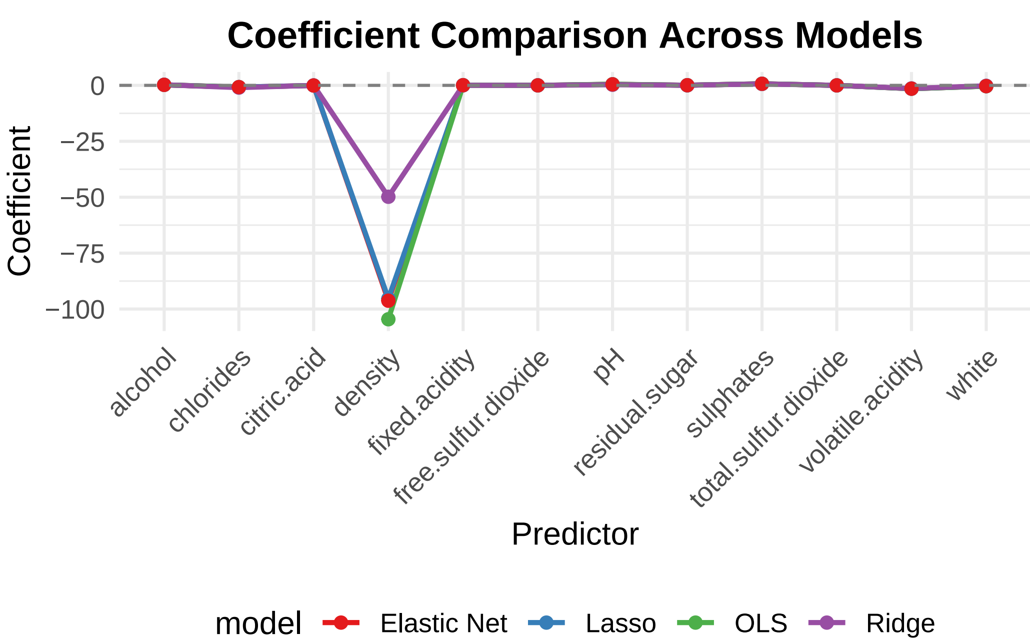

Figure 11: Coefficient comparison across OLS and shrinkage methods

The coefficient comparison helps show how regularization changes effect sizes. Ridge tends to shrink all coefficients, while Lasso and Elastic Net may also reduce some of them to zero or near zero.

6.4 Predictions Comparison

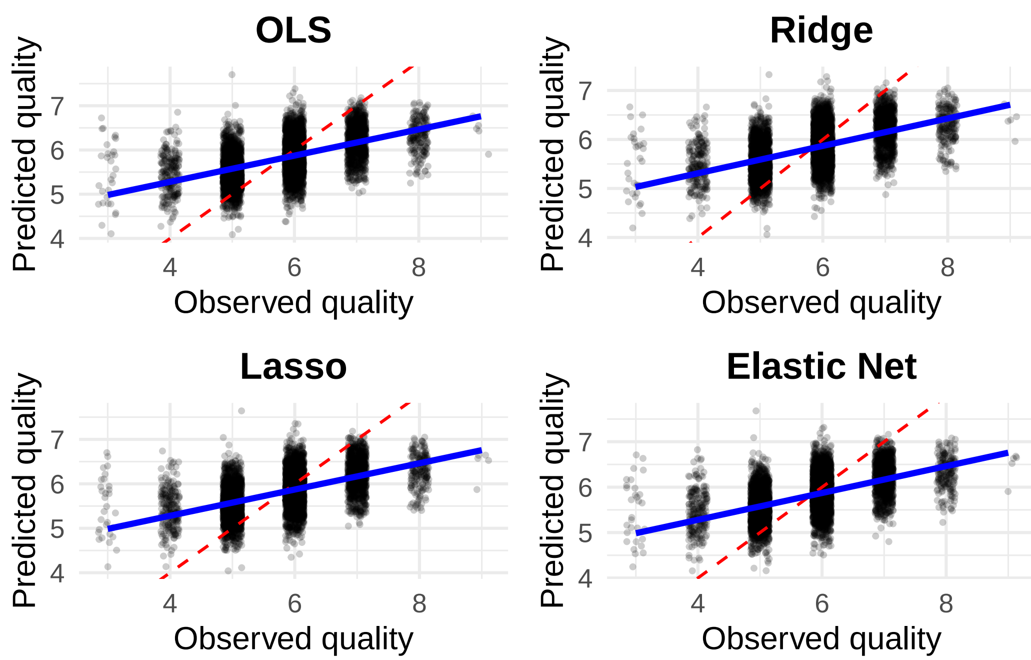

Figure 12: Predicted versus observed wine quality across OLS and shrinkage methods

These panels show how tightly each model tracks the observed quality scores. They are useful because they visualize not only average fit but also the spread of prediction errors.

7 Key Findings and Recommendations

7.1 Summary of Results

Table 20: High-level summary of the main model comparison findings

finding	value
Best model by R-squared	Elastic Net
Best model by RMSE	Elastic Net

finding	value
Predictors retained by AIC-selected OLS	11
Non-zero predictors retained by Lasso	12
Non-zero predictors retained by Elastic Net	12

Table 21: Predictors with the largest average absolute coefficients across models

term	average_absolute_coefficient
density	86.5140
volatile.acidity	1.4556
chlorides	0.8351
sulphates	0.6962
pH	0.4271

The overall comparison shows whether shrinkage improves predictive performance and which variables remain influential across modeling frameworks. This is especially helpful when I want to distinguish stable signals from coefficients that are sensitive to model choice.

7.2 Practical Recommendations

The model results suggest that some physicochemical variables carry more consistent weight than others across OLS and regularized regression. Predictors such as alcohol, volatile acidity, sulphates, and density often remain among the most influential variables, although their exact magnitudes vary by method. In practical terms, that means the report points to a small set of recurring quality-related measurements rather than a completely unstable predictor landscape.

For predictive modeling, the strongest recommendation is to prefer the model that balances fit, stability, and interpretability rather than focusing on only one metric. If the best-performing shrinkage model produces slightly better prediction with a cleaner coefficient pattern, it is a strong candidate for final reporting. If interpretability is the main priority, the AIC-selected OLS model remains useful because the coefficients are straightforward to explain and the assumptions have been checked directly.

For a wine-production interpretation, the results are most useful as directional guidance rather than as a mechanical recipe. Variables that repeatedly show negative associations with quality may identify chemical patterns worth controlling more tightly, while variables with repeated positive associations may indicate features linked to stronger quality ratings in this dataset.

7.3 Model Selection Guide

Table 22: Guide to when each modeling approach is most useful

method	use_when
OLS	You want interpretable coefficients, formal diagnostics, and a classical regression benchmark.
Ridge	You expect multicollinearity and want to keep all predictors while stabilizing coefficient estimates.
Lasso	You want automatic variable selection and a sparser model.
Elastic Net	You want both shrinkage and variable selection when predictors are correlated.

This guide places the model comparison into practical context. Rather than treating one method as universally best, it links each method to the kind of analytical goal it serves most effectively.

8 Appendix

8.1 Session Information

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.2

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] gridExtra_2.3   glmnet_4.1-10   Matrix_1.7-4    lmtest_0.9-40
```

```
 [5] zoo_1.8-14      car_3.1-3       carData_3.0-5   broom_1.0.8
 [9] knitr_1.50      lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
[13] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
[17] tibble_3.3.0    ggplot2_4.0.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] generics_0.1.4     shape_1.4.6.1      stringi_1.8.7      lattice_0.22-7
 [5] hms_1.1.3          digest_0.6.37      magrittr_2.0.3     evaluate_1.0.4
 [9] grid_4.5.2         timechange_0.3.0   RColorBrewer_1.1-3 iterators_1.0.14
[13] fastmap_1.2.0      foreach_1.5.2      jsonlite_2.0.0     backports_1.5.0
[17] Formula_1.2-5      tinytex_0.59       survival_3.8-3     mgcv_1.9-3
[21] scales_1.4.0       codetools_0.2-20   abind_1.4-8        cli_3.6.5
[25] rlang_1.1.6        splines_4.5.2      withr_3.0.2        yaml_2.3.12
[29] tools_4.5.2        tzdb_0.5.0         vctrs_0.6.5        R6_2.6.1
[33] lifecycle_1.0.4    pkgconfig_2.0.3    pillar_1.11.0      gtable_0.3.6
[37] Rcpp_1.1.0         glue_1.8.0         xfun_0.52          tidyselect_1.2.1
[41] rstudioapi_0.17.1  farver_2.1.2       nlme_3.1-168       htmltools_0.5.8.1
[45] labeling_0.4.3     rmarkdown_2.29     compiler_4.5.2     S7_0.2.0
```

9 References